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A novel identity-based multi-signature scheme over NTRU lattices ☆

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ABSTRACT

Multi-signature is an important technology to compress multiple signatures on the common message into a compact one, thereby reducing the consumption of storage space and transmission bandwidth. Such a cryptographic primitive is widely used in financial applications such as blockchain which requires multiple keys to authorize a transaction. With the advent of quantum computers, traditional multi-signature schemes may no longer be secure as their underlying security assumptions (e.g. RSA or discrete logarithm problems) may not hold anymore. In this paper, we propose a novel identity-based multi-signature (IBMS) scheme over NTRU lattices, which is secure against the attacks of quantum computers. To the best of our knowledge, it is the first lattice-based IBMS scheme. We show that our scheme is provably secure in the random oracle model based on the ring version of the short integer solution assumption (Ring-SIS). Compared with the closely related works in the literature, our scheme enables the signer to select its system identity (such as email address, physical IP address, and etc.) as the public key, which effectively alleviates the certificate management problem in the PKI setting, and takes the advantage of discrete Gaussian distribution instead of uniform distribution to generate secret signing keys and multi-signatures. Besides, our scheme does not require all the system users to setup a trusted common string, which further simplifies the deployment of our scheme in practice.
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1. Introduction

1.1. Multi-signature

In 1976, Diffie and Hellman [10] firstly introduced the concept of digital signature. Since then, there have been a bunch of researches on digital signature. Digital signature is a key technology for simulating physical/hand-written signatures in the network environment. Generally, digital signature includes two algorithms, a signing algorithm used to sign on the messages, and a verification algorithm for verifying the validity of signatures. The message sender uses its own secret key to run the signing algorithm to generate a digital signature on the message. Anyone could run the public verification

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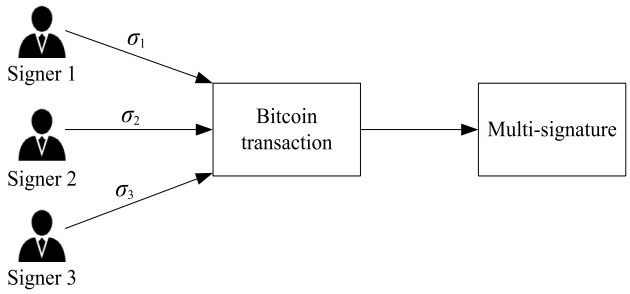

Fig. 1. Multi-signature in bitcoin transaction.

algorithm to check the validity of the signature (and the message). Validity of the signature ensures the authenticity of the message. As an important authentication technology, digital signature plays an indispensable role in the fields of identity authentication and information security [50].

With various application requirements, there are many variants of digital signature. In practice, it is often necessary for several units or departments to sign on the same document separately for this document to take effect. Multi-signature technology is a way to simulate this scenario in the network environment. In 1983, Itakura and Nakamura [25] firstly proposed the notion of *multi-signature*, in which a group of signers with public keys $\{pk_1, \cdots, pk_n\}$ agree to jointly construct a multi-signature on the common message m. In this way, the resulting multi-signature size could be much smaller compared with a bundled signature using n single signatures [14], thereby significantly reducing the consumption of storage space and transmission bandwidth. According to whether to sign the message in order, multi-signature could be divided into two categories, *broadcasting* multi-signature and *sequential* multi-signature. In the former, each signer signs the message individually and broadcasts its signature to others. Anyone collecting all the individual signatures could combine them into a single multi-signature. While in the latter, all the signers sign the message sequentially. After receiving the signature σ_{i-1} from the $(i-1)$-th signer, the i-th signer integrates its signature on the message into σ_{i-1} and sends the updated signature σ_i to the next signer. According to [7], in standard multi-signature schemes, security is commonly defined through an experiment with a single honest "target" signer, while viewing all other signers as corrupted. Security requires that it should be infeasible to forge a multi-signature on a new message involving the target signer, after seeing polynomially many signatures on messages of its choice.

Multi-signature has extensive applications, including blockchain bitcoin transactions [5,26,35], electronic contract signing [20,28], electronic medical record [16,29] and etc. Below we take the blockchain transaction as an example to illustrate the application of multi-signature.

Application in blockchain: Since the proposal of multi-signature, many multi-signature schemes and applications have emerged. Recently, the application of multi-signature has ushered in a new development trend with the rise of blockchain technology, because multi-signature can be applied to the blockchain technology and efficiently reduces the consumption of storage space and transmission bandwidth on the blocks. Concretely, more complicated transactions are supported in the bitcoin blockchain as the bitcoin network is developing rapidly in recent years, such as M-out-of-N transactions which require not all but M valid signatures to take effect and store these signatures to the blockchain [35]. Briefly speaking, if M of the N transactors agree to this transaction (providing M valid signatures), the transaction will be validated. In this case, it is significant to compress multiple signatures into a joint one for the sake of the storage in blockchains. Take a 2-out-of-3 transaction for instance, as shown in Fig. 1. A transaction can be executed only if 2 of the 3 signers provide valid signatures.

Multi-signature has a number of applications in Bitcoin transactions and wallets. For example, we can use multi-signature to (1) setup a joint account and divide up the responsibility for the possession of bitcoins among multiple owners, (2) to increase security of the bitcoin wallet and make it more difficult for the wallet to be compromised, and (3) to make an M-out-of-N backup so that the loss of a single key does not lead to the loss of the wallet.

Traditional public key cryptosystems work in the public key infrastructure (PKI). In order to bind the public key of a user with its identity, a trusted third-party called *certificate authority* (CA) issues digital certificates to users. However, the management of certificates, such as issuance, storage, verification, revocation and etc, is a complex work. As the number of users increases, the cost of certificate management grows. To solve this problem, Shamir introduced the notion of *identity-based cryptography* (IBC) in 1984 [43], in which each user selects their identity information (such as email address, physical IP address and etc.) as the public key, and obtains the corresponding secret key from a trusted party called *private key generator* (PKG). Anyone could encrypt a message using the receiver's identity even before the receiver gets its secret key from the PKG. IBC alleviates the certificate management problem effectively. Since then, IBC has attracted a lot of researchers' attention and a bunch of identity-based encryption schemes and signature schemes have been proposed. We study identity-based multi-signature in this paper.

Post-quantum security: Along with the development of quantum computing technology, traditional cryptographic schemes may no longer be secure as their underlying security assumptions (e.g. RSA or discrete logarithm problems) may not hold

anymore. In 1994, Shor proposed efficient algorithms to solve discrete logarithm problem and integer factorization problem on a quantum computer, which run in time polynomial in the problem size [44]. Subsequently, Grover presented a quantum mechanical search algorithm which is polynomially faster than classical algorithms [15]. As a result, with the help of Shor's algorithms and Grover's algorithm, both traditional difficult problems could be solved in polynomial time on a quantum computer. Therefore, traditional digital signature schemes with security based on the intractability of classical hard problems are facing serious security threats. Since then, some new alternatives have been proposed for solving this vexing situation, among which lattice-based cryptography is the most anticipated. Compared with the traditional intractability assumptions, lattice problems are considered to be able to resist the attacks of quantum computing. There is no polynomial-time algorithm which could solve lattice problems even on a quantum computer up to now. Therefore, in this paper we focus on the construction of identity-based multi-signature schemes based on hard lattice problems to withstand quantum attacks, which is especially significant for financial applications.

1.2. Related works

In 1983, Itakura and Nakamura [25] firstly presented the definition of multi-signature. Subsequently, lots of works had been proposed including multi-signature schemes [30,32], but there were some flaws in the security proofs or security models of these schemes. Actually, only the schemes proposed by Micali et al. [38] and Okamoto et al. [40] meet the security requirements. Furthermore, security of the former scheme is stronger than the latter one, since it does take rogue-key attacks [7] into account. However, we need know the set of signers in advance before applying the scheme [38]. In 2006, Bellare and Neven provided a three-round provably secure multi-signature scheme in the plain public key model [7]. In this model, it only requires that signers in the signing group should have a public key. Note that the signing process of [7] is an interactive protocol, in which each member in the signing group has to collaborate before outputting a multi-signature. Afterwards, Bagherzandi et al. [3] and Ma et al. [36] improved the scheme [7] and presented two-round multi-signature schemes in 2008 and 2010, respectively. Subsequently, Syta et al. [46] presented a highly scalable multi-signature scheme in 2016. The first multi-signature scheme supporting key aggregation was proposed by Maxwell et al. [37] in 2018, which is proved to be secure in the plain public key model. Simultaneously, new pairing-based multi-signature schemes were presented by Boneh et al. [8] which support not only key aggregation but also batch verification. To solve the certificate management problem in PKI-based schemes, lots of identity-based multi-signature schemes have been proposed, such as [2,4,6,24,49].

However, the schemes mentioned above may no longer be secure as their underlying security assumptions (e.g. RSA or discrete logarithm problems) may not hold anymore with the advent of quantum computers. Hence, some lattice-based multi-signature schemes were introduced. In 2013, a broadcast multi-signature scheme and a sequential multi-signature scheme were proposed by Kong et al. [27]. The multi-signature size of the former scheme is linearly related to the number of signers while that of the latter is constant and is independent of the number of signers. In 2016, Peng et al. [41] proposed a lattice-based multi-signature scheme, which combines the individual signatures with an NIWI proof [39] to form the multi-signature. However, the length of the multi-signature has a linear relationship with the number of signers. Subsequently, a lattice-based linear homomorphic multi-signature scheme was proposed by Choi and Kim [9] in 2016, in which the role of a dealer was used. However, the scheme has a cumbersome reset process when other members join. In the same year, Bansarkhani and Sturm [5] proposed a provably secure lattice-based broadcast multi-signature scheme, whose signature length is close to a single signature. Unfortunately, the efficiency of their signature and verification algorithms is linear with the number of signers. In 2019, both Ma and Jiang [35] and Fukumitsu and Hasegawa [12] presented multi-signature schemes over lattice based on the scheme in [5], respectively. Recently, Kansal et al. [26] constructed a lattice-based multi-signature scheme that supports public key aggregation in the plain public key model based on [5] as well. Notice that the schemes [5,12,26,35] actually work in the common reference string model. All the system users have to share a trusted common string, which could be generated either by a trusted party or by all the users interactively via a secure coin tossing protocol. It is assumed that no user could have the corresponding trapdoor related to the common string, which is not easy to implement in practice. In the recent years, some works about lattice have been proposed such as [17,18,42,47], to introduce the optimization on lattices and some new research directions.

To the best of our knowledge, all the lattice-based multi-signature schemes work in the public-key infrastructure and suffer from the complex certificate management problem. There is no identity-based multi-signature (IBMS) scheme with security against quantum attacks up to now.

1.3. Our contributions

In this paper we solve the aforementioned problem by constructing an IBMS scheme based on hard lattice problem. Concretely, we make the following contributions in the paper.

1. We propose an IBMS scheme over NTRU lattices, which to the best of our knowledge, is the first lattice-based IBMS scheme in the literature. In our scheme each signer could simply use its identity information as its public key, thus solving the cumbersome certificate management problem. The multi-signature generation process is a six-move inter-

active protocol among all the signers, while the signature verification is non-interactive, which could be done by each individual verifier.
2. We prove our IBMS scheme to be secure in the random oracle model based on the ring version of the *short integer solution* assumption (Ring-SIS) which is a well studied problem and is commonly used in constructing lattice-based cryptographic schemes.
3. Compared with the RIBS scheme [23] which is an identity-based signature (not multi-signature) scheme over lattice, our IBMS scheme supports the multi-signature functionality. Signature size in our scheme is much smaller than a bundled signature consisting of N single signatures (of RIBS scheme), as we perform an accumulation operation on N signatures instead of simply concatenating them. Compared with the existing multi-signature schemes [5,12,26,35] which work in the PKI setting, our scheme implements the identity-based functionality, and takes advantage of discrete Gaussian distribution instead of uniform distribution to generate secret signing keys and multi-signatures.
4. Besides, our scheme does not require all the system users to share a trusted common string, which further simplifies the deployment of our scheme in practice.

1.4. Organization

The rest of this paper is organized as follows. Some preliminaries are introduced in Section 2. Section 3 presents the syntax and security model of IBMS scheme. In Section 4, our IBMS scheme over NTRU lattices is proposed, which is proved to be secure in Section 5. Performance analysis and comparisons are provided in Section 6, illustrating the advantages of our IBMS scheme. Finally, conclusions are given in Section 7.

2. Preliminaries

2.1. Notations

Throughout the paper, we let n be a positive power-of-two integer and q be a prime convergent to 1 modulo $2n$. $\mathbb{R}$ stands for the set of real numbers while $\mathbb{Z}$ represents the set of integers. $\mathbb{Z}_q$ defines the set of integers in the range $[-(q-1)/2, (q-1)/2]$. $\|\mathbf{v}\|_\infty$ denotes the infinity norm of a vector $\mathbf{v}$ whose Euclidean norm can be represented by $\|\mathbf{v}\|$ (the subscript omitted). For a set $\mathbf{B} = \{\mathbf{b}_1, \mathbf{b}_2, \cdots, \mathbf{b}_n\}$ of n linearly independent vectors, we take $\widetilde{\mathbf{B}} = \{\widetilde{\mathbf{b}}_1, \widetilde{\mathbf{b}}_2, \cdots, \widetilde{\mathbf{b}}_n\}$ to denote its Gram-Schmidt orthogonalization. We assume that $\mathcal{R}$ represents the ring $\mathbb{Z}[x]/(x^n+1)$ and $\mathcal{R}_q$ denotes the ring $\mathbb{Z}_q[x]/(x^n+1)$, where the elements can be denoted as polynomials with coefficients in $\mathbb{Z}_q$ and degree $n-1$. Let $f = \sum_{i=0}^{n-1} f_i x^i$ and $g = \sum_{i=0}^{n-1} g_i x^i$ be two polynomials in $\mathcal{R}_q$. For convenience, we take a vector $[f_0, f_1, ..., f_{n-1}]$ to represent a polynomial f. There are two mathematical operations between f and g as follows. One is addition $f+g$ in $\mathcal{R}_q$, the other is multiplication $f \cdot g$ in $\mathcal{R}_q$. Note that the polynomial coefficients of both operation results are reduced modulo q.

$$\begin{aligned} f+g &= \sum_{i=0}^{n-1} (f_i + g_i)\, x^i \pmod q \\ f \cdot g &= \sum_{k=0}^{n-1} \left(\sum_{i+j=k \pmod n} f_i g_j \right) x^k \pmod q \end{aligned} \tag{1}$$

2.2. Lattice

Lattice is a set of discrete points with periodic structure [1]. Let $\mathbf{b}_1, \mathbf{b}_2, \cdots, \mathbf{b}_n$ stand for n linearly independent vectors in $\mathbb{R}^n$, which generates an (n-dimensional) lattice $\mathbf{\Lambda}$, denoted by $\mathcal{L}(\mathbf{b}_1, \mathbf{b}_2, \cdots, \mathbf{b}_n)$, representing the set $\{\sum_{i=1}^{n} x_i \mathbf{b}_i : x_i \in \mathbb{Z}\}$. The set $\mathbf{B} = \{\mathbf{b}_1, \mathbf{b}_2, \cdots, \mathbf{b}_n\}$ is a basis of lattice $\mathbf{\Lambda}$. The same lattice can be represented by different lattice bases.

2.3. Anticirculant matrices

Definition 1 *(Anticirculant matrix).* An N-dimensional anticirculant matrix of $f = \sum_{i=0}^{N-1} f_i x^i$ is defined as follows.

$$A_N(f) = \begin{pmatrix} (f) \\ (x \cdot f) \\ \vdots \\ (x^{N-1} \cdot f) \end{pmatrix} = \begin{pmatrix} f_0 & f_1 & \cdots & f_{N-1} \\ -f_{N-1} & f_0 & \cdots & f_{N-2} \\ \vdots & \vdots & \vdots & \vdots \\ -f_1 & -f_2 & \cdots & f_0 \end{pmatrix} \tag{2}$$

For convenience, the subscript N will be left out and we will take $A(f)$ to represent the anticirculant matrix in this paper. According to [11], anticirculant matrices possess the following property.

Lemma 1 *([11]). For polynomials $f, g \in \mathcal{R}_q$, $A(f) \times A(g) = A(f \cdot g)$ and $A(f) + A(g) = A(f+g)$.*

2.4. NTRU lattice

NTRU lattice is a special kind of convolution modular lattices. NTRU [22] is a well-known cryptosystem based on NTRU lattice in the literature. Below, we briefly introduce the concept of NTRU lattice used in this paper.

Definition 2 *(NTRU lattice).* Suppose that n is a power-of-two integer, q is a positive integer, and $h = g \cdot f^{-1} \bmod q$ where $f, g \in \mathcal{R}_q$. The NTRU lattice is

$$\mathbf{\Lambda}_{h,q} = \left\{ (u, v) \in \mathcal{R}^2 \middle| u + h \cdot v = 0 \bmod q \right\}. \tag{3}$$

Indeed, $\mathbf{\Lambda}_{h,q}$ is a full-rank lattice of $\mathbb{Z}^{2n}$ generated by the matrix

$$\mathbf{A}_{h,q} = \begin{pmatrix} -A(h) & I_n \\ qI_n & O_n \end{pmatrix}, \tag{4}$$

where O_n and I_n are the $n \times n$ null matrix and $n \times n$ unit matrix, respectively.

Obviously, a single polynomial $h \in \mathcal{R}_q$ can uniquely define the basis $\mathbf{A}_{h,q}$, which is storage-efficient since we only need store a single polynomial h instead of the whole matrix. However, the basis $\mathbf{A}_{h,q}$ will suffer a very large orthogonal defect if h is uniformly distributed in $\mathcal{R}_q$, which makes it more difficult for solving lattice problems. To make up for this, Hoffstein et al. [21] introduced another more suitable basis

$$\mathbf{B}_{f,g} = \begin{pmatrix} A(g) & -A(f) \\ A(G) & -A(F) \end{pmatrix}, \tag{5}$$

where $F, G \in \mathcal{R}_q$ and $f \cdot G - g \cdot F = q$. In addition, we can obtain the tuple (F, G) efficiently according to [11]. Furthermore, a short basis for $\mathbf{\Lambda}_{h,q}$ is provided according to the following lemma.

Lemma 2 *([11]). Let $f, g, F, G \in \mathcal{R}_q$ satisfying $h = g \cdot f^{-1} \bmod q$ and $f \cdot G - g \cdot F = q$. Then both $\mathbf{A}_{h,q}$ and $\mathbf{B}_{f,g}$ are bases of the NTRU lattice $\mathbf{\Lambda}_{h,q}$ and $\|\mathbf{B}_{f,g}\| \le \|\mathbf{A}_{h,q}\|$.*

In 2014, Ducas et al. [11] instantiated the trapdoor generation algorithm in GPV scheme [13] taking advantage of a specific distribution over NTRU lattices. In this paper, we construct our trapdoor generation algorithm according to [11] as follows.

Lemma 3 *([11]). For a prime q, a power-of-two integer n, and $\sigma_f = 1.17\sqrt{\frac{q}{2n}}$, there exists a probabilistic polynomial-time (PPT) algorithm* **TrapGen**(q, n) *to generate a tuple (f, g) and output $h = g \cdot f^{-1} \bmod q$ statistically close to uniform in $\mathcal{R}_q$, and a short basis $\mathbf{B}_{f,g}$ of $\mathbf{\Lambda}_{h,q}$.*

2.5. Gaussian distribution and rejection sampling

Definition 3 *(Gaussian distribution).* With standard deviation $s > 0$ and center $\mathbf{c} \in \mathbb{R}^n$, the continuous Gaussian distribution over $\mathbb{R}^n$ can be defined by the following function, where $\mathbf{v} \in \mathbb{R}^n$.

$$\rho_{s,\mathbf{c}}(\mathbf{v}) = \left(\frac{1}{\sqrt{2\pi}s} \right)^n e^{\frac{-\|\mathbf{v}-\mathbf{c}\|^2}{2s^2}}. \tag{6}$$

Let $\rho_{s,\mathbf{c}}(\mathbf{\Lambda})$ denote $\sum_{\mathbf{v}\in\mathbf{\Lambda}} \rho_{s,\mathbf{c}}(\mathbf{v})$ for any lattice $\mathbf{\Lambda} \in \mathbb{R}^n$.

Definition 4 *(Discrete Gaussian distribution).* With standard deviation $s > 0$ and center $\mathbf{c} \in \mathbb{R}^n$, the discrete Gaussian distribution over $\mathbb{R}^n$ can be defined by the following function, where $\mathbf{v} \in \mathbf{\Lambda}$.

$$\mathcal{D}_{\mathbf{\Lambda},s,\mathbf{c}}(\mathbf{v}) = \rho_{s,\mathbf{c}}(\mathbf{v}) / \rho_{s,\mathbf{c}}(\mathbf{\Lambda}). \tag{7}$$

When $\mathbf{c} = \mathbf{0}$, we can drop the script $\mathbf{c}$ and abbreviate $\mathcal{D}_{\mathbf{\Lambda},s,\mathbf{0}}$ and $\rho_{s,\mathbf{0}}$ as $\mathcal{D}_{\mathbf{\Lambda},s}$ and ρ_s, respectively. To simplify, we focus on integer lattices. Tian et al. [48] proposed two basic properties of discrete Gaussian distributions $\mathcal{D}_{\mathbb{Z}^n,\sigma}$ over the n-dimensional integer lattice with standard deviation $\sigma > 0$ as follows.

Lemma 4 *([48]). For any positive integer n and standard deviation $\sigma > 0$, we have the following properties:*

(1) $\Pr\left[v \leftarrow \mathcal{D}_{\mathbb{Z},\sigma} : |v| > 12\sigma\right] < 2^{-100}$.
(2) $\Pr\left[\mathbf{v} \leftarrow \mathcal{D}_{\mathbb{Z}^n,\sigma} : \|\mathbf{v}\| > 2\sigma\sqrt{n}\right] < 2^{-n}$.

Furthermore, Lyubashevsky [33] proposed the following properties of $\mathcal{D}_{\mathbb{Z}^n,\sigma,\mathbf{c}}$, the discrete Gaussian distribution over the n-dimensional integer lattice with standard deviation $\sigma > 0$ at center $\mathbf{c}$.

Lemma 5 *([33]). For any $\mathbf{c} \in \mathbb{Z}^n$ and $\alpha > 0$, if $\sigma = \omega(\|\mathbf{c}\|\sqrt{\log n})$, we have*

$$\Pr\left[\mathbf{v} \leftarrow \mathcal{D}_{\mathbb{Z}^n,\sigma} : \mathcal{D}_{\mathbb{Z}^n,\sigma}(\mathbf{v})/\mathcal{D}_{\mathbb{Z}^n,\sigma,\mathbf{c}}(\mathbf{v}) = O(1)\right] = 1 - 2^{-\omega(\log n)}, \tag{8}$$

and specifically, if $\sigma = \alpha\|\mathbf{c}\|$, we have

$$\Pr\left[\mathbf{v} \leftarrow \mathcal{D}_{\mathbb{Z}^n,\sigma} : \mathcal{D}_{\mathbb{Z}^n,\sigma}(\mathbf{v})/\mathcal{D}_{\mathbb{Z}^n,\sigma,\mathbf{c}}(\mathbf{v}) < e^{12/\alpha+1/(2\alpha^2)}\right] > 1 - 2^{-100}. \tag{9}$$

According to [48], we can obtain a real $M \approx e$ satisfying $\mathcal{D}_{\mathbb{Z}^n,\sigma}(\mathbf{v}) \le M\mathcal{D}_{\mathbb{Z}^n,\sigma,\mathbf{c}}(\mathbf{v})$ for all $\mathbf{v} \in \mathcal{D}_{\mathbb{Z}^n,\sigma}$.

The rejection sampling technique is often taken into lattice-based signature schemes so as to make the signature distribution independent of the secret signing key. When a signer wants to sign a message using its signing key S_{ID} associated with the signer's ID, it can choose a vector $\mathbf{y} \in \mathcal{D}_{\mathbb{Z}^n,\sigma}$ and compute a candidate signature $\mathbf{z} = S_{ID} \cdot \mathbf{c} + \mathbf{y}$, where $\mathbf{c}$ represents the message's hash value. Assume that $\mathscr{F}$ is the target distribution (independent of the signing key S_{ID}) of the output signature $\mathbf{z}$. Furthermore, suppose there exists a probability distribution $\mathscr{G}$ and a real $M > 0$ satisfying for all $\mathbf{v}$, $\mathscr{F}(\mathbf{v}) \le M \cdot \mathscr{G}(\mathbf{v})$ always holds, then the signing algorithm successfully outputs the candidate signature $\mathbf{z}$ with probability $\min[\mathscr{F}(\mathbf{z})/M \cdot \mathscr{G}(\mathbf{z}), 1]$, which means it outputs signature $\mathbf{z}$ if $\mathscr{F}(\mathbf{z}) \le M \cdot \mathscr{G}(\mathbf{z})$ holds and rejects it otherwise. Additionally, M is the expected repetition times needed to successfully output a signature.

2.6. Sampling algorithm

Definition 5 *(Sampling algorithm).* Assume that q is a prime, $\mathbf{B}_{f,g}$ is a short basis of the 2n-dimensional lattice $\mathbf{\Lambda}_{h,q}$ and $t \in \mathcal{R}_q$. If $0 < \varepsilon < 1$ and the Gaussian parameter $s \ge \|\widetilde{\mathbf{B}_{f,g}}\|\omega(\sqrt{\log 2n})$, then for any $\mathbf{u} = (t, \mathbf{0}) \in \mathbb{Z}_q^{2n}$, we have the following conclusions [11,13,31]:

(1) $\Pr\left[\mathbf{x} \leftarrow \mathcal{D}_{\mathbf{\Lambda}_{h,q},s,\mathbf{u}} : \|\mathbf{x} - \mathbf{u}\| > s \cdot \sqrt{2n}\right] \le (1+\varepsilon)/(1-\varepsilon) \cdot 2^{-2n}$; and
(2) there is a PPT algorithm **SampleGau**$(\mathbf{B}_{f,g}, s, \mathbf{u})$ that can output a sample $(s_1, s_2) \in \mathcal{R}_q^2$ satisfying $s_1 + h \cdot s_2 = t$ and $\|(s_1, s_2)\| \le s \cdot \sqrt{2n}$.

2.7. Hardness assumption

Below, we present the definition of ring short integer solution (Ring-SIS) problem [35], which is the underlying security assumption of our IBMS scheme.

Definition 6 *(Ring-SIS$_{m,q,\beta}$ problem).* Given $\mathbf{a} = (a_1, a_2, \cdots, a_m)^T \in \mathcal{R}_q^m$, a vector of m chosen uniformly polynomials, the associated Ring-SIS$_{m,q,\beta}$ problem is to find a nonzero vector of small polynomials $\mathbf{x} = (x_1, x_2, \cdots, x_m)^T \in \mathcal{R}^m$ satisfying $0 < \|\mathbf{x}\| \le \beta$ and $\mathbf{a}^T\mathbf{x} = \sum_{i=1}^m a_i \cdot x_i = 0 \bmod q$.

According to [45], we can choose $h = g \cdot f^{-1} \bmod q$ statistically close to uniform in $\mathcal{R}_q$ as an NTRU public key, then the Ring-SIS$_{m,q,\beta}$ problem on the NTRU lattices is to find a nonzero vector $(\mathbf{x}_1, \mathbf{x}_2)$ satisfying $0 < \|(\mathbf{x}_1, \mathbf{x}_2)\| \le \beta$ and $\mathbf{x}_1 + h \cdot \mathbf{x}_2 = 0 \bmod q$, which is believed to be hard.

3. Definition and security model of IBMS

3.1. Definition of IBMS

Suppose there are N signers in the signing group with identity set $\mathbf{ID} = \{ID_1, ID_2, \cdots, ID_N\}$. An IBMS scheme consists of four algorithms (**MSetup**, **MExtract**, **MSign**, **MVerify**) defined as follows:

- **MSetup**(1^n): The setup algorithm takes a security parameter 1^n as input, and outputs the system secret key S_{PKG} (secret key of the private key generator (PKG)) and public parameters *params*.
- **MExtract**$(ID_i, S_{PKG}, params)$: Run by the PKG, the extraction algorithm aims to generate signing keys for signers according to their identities. It takes an identity ID_i of signer i, the PKG's secret key S_{PKG} and the public parameters *params* as input, and outputs a signing key S_i for the signer i.

- **MSign**(**ID**, m, *params*): The signing algorithm takes an identity set $\mathbf{ID} = \{ID_1, ID_2, \cdots, ID_N\}$, a message m and the public parameters *params* as input, and outputs a collective multi-signature θ or a symbol $\perp$ to denote failure. In some IBMS schemes, the multi-signature generation algorithm is an interactive protocol that requires all the signers to interact with each other before outputting the final multi-signature. In such a protocol, the signing keys of signers are seen as their local private input.
- **MVerify**(**ID**, θ, m, *params*): The verification algorithm takes an identity set $\mathbf{ID} = \{ID_1, ID_2, \cdots, ID_N\}$, a multi-signature θ, a message m and the public parameters *params* as input, and outputs 1 if the multi-signature θ is valid and 0 otherwise.

Briefly speaking, if all signers follow the **MSign** protocol honestly to sign on the common message m, they would generate a valid multi-signature θ satisfying that the verification algorithm **MVerify** outputs 1 while taking **ID**, θ, m, and *params* as input, which is required for the correctness of an IBMS scheme.

3.2. Security model of IBMS

It should be infeasible to forge a valid multi-signature even if there is only one honest signer, which is required for the security of IBMS. According to [7], we suppose there exists only one honest signer with the identity ID^* in the signing group. Interacting with the honest signer, the adversary $\mathcal{A}$ can start an **MSign** algorithm with identity set $\mathbf{ID} = \{ID_1, ID_2, \cdots, ID_N\}$ and a message m, where $ID^* \in \mathbf{ID}$. The security model of IBMS is described by the following game.

(1) **Setup**. The challenger C runs the **MSetup** algorithm to set S_{PKG} and *params*. C then keeps S_{PKG} secret and sends *params* to the adversary $\mathcal{A}$.
(2) **Queries**. $\mathcal{A}$ can adaptively issue polynomially many queries as follows.
  (a) **MExtract-query**. When $\mathcal{A}$ queries a signer's secret key with ID_i, C runs the **MExtract** algorithm to generate the secret key S_i and returns S_i to $\mathcal{A}$.
  (b) **MSign-query**. When $\mathcal{A}$ queries a multi-signature with an identity set $\mathbf{ID} = \{ID_1, ID_2, \cdots, ID_N\}$ and a message m, C runs the **MSign** algorithm to obtain a multi-signature θ and returns θ to $\mathcal{A}$.
(3) **Forgery**. $\mathcal{A}$ outputs a multi-signature tuple $\left(m', ID^*, \mathbf{ID}', \theta'\right)$. If the tuple satisfies the following conditions, we say $\mathcal{A}$ wins the game:
  (a) **MVerify**$(\mathbf{ID}', \theta', m', params) = 1$;
  (b) $ID^* \in \mathbf{ID}'$, and ID^* has not been issued during the **MExtract-query** phase; and
  (c) the tuple $\left(\mathbf{ID}', m'\right)$ has not been issued during the **MSign-query** phase.

Definition 7 *(EUF-ID-CMA).* An *IBMS* scheme is existentially unforgeable under chosen-identity and chosen-message attacks if for any PPT adversary $\mathcal{A}$ the probability that $\mathcal{A}$ wins the above game is negligible.

If the adversary is required to submit the challenge identity ID^* before seeing any public system parameter, then we say that an IBMS scheme is existentially unforgeable under *selective*-identity and chosen-message attacks (*EUF-SID-CMA* secure) if no PPT adversary can win the modified game with non-negligible probability.

4. Our IBMS scheme over NTRU lattices

Below, we describe the algorithms of our IBMS scheme over NTRU lattices.

- **MSetup**. Given the system parameter 1^n, the PKG chooses two real numbers $s, \sigma > 0$ and a prime q, and runs the algorithm **TrapGen**(q, n) to generate a tuple (f, g) satisfying $\|f\| \le s\sqrt{n}$, $\|g\| \le s\sqrt{n}$ and $h = g \cdot f^{-1} \bmod q$ as well as a short basis

$$\mathbf{B}_{f,g} = \begin{pmatrix} A(g) & -A(f) \\ A(G) & -A(F) \end{pmatrix} \in \mathbb{Z}_q^{2n \times 2n} \text{ of } \mathbf{\Lambda}_{h,q}. \tag{10}$$

  In addition, the system secret key S_{PKG} and public parameters *params* are generated by PKG as follows:
  (a) Select three hash functions $H_0 : \{0,1\}^* \to \mathbb{Z}_q^n$, $H_1 : \mathbb{Z}_q^n \to \mathbb{Z}_q^n$, and $H_2 : \{0,1\}^* \to \{\mathbf{e} | \mathbf{e} \in \{-1, 0, 1\}^n, \|\mathbf{e}\|_1 \le \lambda\}$ in which the hash results $\mathbf{e}$ stand for polynomials whose coefficients are all zero except for λ nonzero components.
  (b) Let $S_{PKG} = \mathbf{B}_{f,g}$ and $params = \langle h, H_0, H_1, H_2 \rangle$, in which h denotes the system public key. Then the PKG outputs the *params* and keeps S_{PKG} in secret.
- **MExtract**. Given $ID_i \in \{0,1\}^*$ of signer i, the PKG calculates $H_0(ID_i)$ and runs the algorithm **SampleGau**$(\mathbf{B}_{f,g}, s, (H_0(ID_i), 0))$ to obtain a sample $(s_{i,1}, s_{i,2})$ satisfying $s_{i,1} + h \cdot s_{i,2} = H_0(ID_i)$ and $\|(s_{i,1}, s_{i,2})\| \le s \cdot \sqrt{2n}$. Finally, the PKG sends the secret key $S_i = (s_{i,1}, s_{i,2})$ to signer i using a secure channel. According to [34], it is quite hard to recover S_i if given h and $H_0(ID_i)$.

MSign

Signer i		Cosigners $j, j \neq i$
(1) $Y_i = (y_{i,1}, y_{i,2}) \leftarrow \mathcal{D}_{\mathbb{Z}^n,\sigma} \times \mathcal{D}_{\mathbb{Z}^n,\sigma}$		
(2) $r_i = y_{i,1} + h \cdot y_{i,2}$		
(3) $d_i = H_1(r_i)$		
	(4) $\xrightarrow{d_i}$	
	(5) $\xleftarrow{d_j}$	
	(6) $\xrightarrow{r_i}$	
	(7) $\xleftarrow{r_j}$	
(8) Check $d_j \stackrel{?}{=} H_1(r_j)$ for all $j \neq i$		
(9) $r = \sum_{j=1}^{N} r_j \bmod q$		
(10) $c_i = H_2(ID_i, r, \mathbf{ID}, m)$		
(11) $z_i = (z_{i,1}, z_{i,2}) \leftarrow S_i c_i + Y_i = ((s_{i,1}c_i + y_{i,1}), (s_{i,2}c_i + y_{i,2}))$ with probability $\min\left[\mathcal{D}_{\mathbb{Z}^n,\sigma}(z_i)/M\mathcal{D}_{\mathbb{Z}^n,\sigma,S_ic_i}(z_i), 1\right]$, $M = O(1)$		
	(12) $\xrightarrow{z_i}$	
	(13) $\xleftarrow{z_j}$	
(14) $z = \sum_{j=1}^{N} z_j$		
(15) Output(z, r)		

Fig. 2. Concrete algorithm **MSign**.

- **MSign**. Given a group of signers with corresponding identity set $\mathbf{ID} = \{ID_1, ID_2, \cdots, ID_N\}$ and a message m, algorithm **MSign** works as follows (see Fig. 2). Our multi-signature generation algorithm is an interactive protocol among all the signers. Without loss of generality, we assume that signer i interacts with other signers j $(j \neq i)$ to obtain the final multi-signature. Firstly, signer i randomly chooses $y_{i,1}, y_{i,2} \in \mathcal{D}_{\mathbb{Z}^n,\sigma}$ and computes $r_i = y_{i,1} + h \cdot y_{i,2}$ and $d_i = H_1(r_i)$. Subsequently, the signer i broadcasts d_i to other cosigners and receives all d_j $(j \neq i)$ from them. It also broadcasts r_i to other cosigners and receives all r_j $(j \neq i)$ from them. Evidently, d_i could be seen as a commitment to r_i, which is used to ensure the correct transmission of r_i. Based on the schemes of [5,7], the signing process in our scheme is a three-round interactive protocol. Signer i sends d_i and r_i to other cosigners separately. If there are some faults when sending d_i, signer i should resend d_i before transmitting r_i. As a result, each signer could check the validity of r_j using d_j and compute $r = \sum_{j=1}^{N} r_j \bmod q$. Subsequently, the signer i queries $H_2(ID_i, r, \mathbf{ID}, m)$ with $\mathbf{ID} = \{ID_1, ID_2, \cdots, ID_N\}$ and obtains $c_i = H_2(ID_i, r, \mathbf{ID}, m)$. It then generates a signature $z_i = S_i c_i + Y_i$ with probability $\min\left[\mathcal{D}_{\mathbb{Z}^n,\sigma}(z_i)/M\mathcal{D}_{\mathbb{Z}^n,\sigma,S_ic_i}(z_i), 1\right]$ where $M = O(1)$, and broadcasts to other cosigners. After receiving all z_j for $j \neq i$ from other cosigners, the signer i could compute $z = \sum_{j=1}^{N} z_j$ and output the final multi-signature (z, r).
- **MVerify**. Given a multi-signature (z, r) on a message m with an identity set $\mathbf{ID} = \{ID_1, ID_2, \cdots, ID_N\}$, algorithm **MVerify** computes $c_i = H_2(ID_i, r, \mathbf{ID}, m)$, and outputs 1 ("accept") if

$$\|z\| \leq 2N \cdot \sigma\sqrt{2n}, \text{ and } z^{(1)} + h \cdot z^{(2)} = r + \sum_{i=1}^{N}(H_0(ID_i) \cdot c_i),$$

  and 0 ("reject") otherwise.

The correctness of our IBMS scheme is shown below.

$$z^{(1)} + h \cdot z^{(2)} = \sum_{i=1}^{N}(z_{i,1} + h \cdot z_{i,2}) = \sum_{i=1}^{N}((s_{i,1}c_i + y_{i,1}) + h \cdot (s_{i,2}c_i + y_{i,2}))$$

$$= \sum_{i=1}^{N}((s_{i,1} + h \cdot s_{i,2})c_i + (y_{i,1} + h \cdot y_{i,2}))$$
$$= \sum_{i=1}^{N}(H_0(ID_i)c_i + r_i)$$
$$= r + \sum_{i=1}^{N}(H_0(ID_i)c_i).$$

5. Security analysis

Theorem 1. *Suppose there exists a polynomial-time forger $\mathcal{F}$, who can make at most ℓ_H queries to random oracles (involving H_0, H_1, H_2), initiate at most ℓ_S signing algorithms with the honest signer including at most N_{max} signers and succeed in providing a forgery of our* IBMS *scheme with probability δ. Then, there exists an algorithm $\mathcal{B}$ (whose time complexity is the same as $\mathcal{F}$) that for a given $\mathbf{A} = (\mathbf{1}, h) \leftarrow \{\mathbf{1}\} \times \mathcal{R}_q$ finds non-zero vectors $\mathbf{u}_1, \mathbf{u}_2 \in \mathbb{Z}_q^n$ satisfying $\mathbf{u}_1 + h \cdot \mathbf{u}_2 = 0$ and $\|\mathbf{u}_1, \mathbf{u}_2\| \le 4N_{max} \cdot \sigma\sqrt{2n} + 2\lambda s\sqrt{2n}$ with probability at least*

$$\left(\frac{1}{2} - 2^{-100}\right)\left(\delta - \frac{3(\ell_H + N \cdot \ell_S + 1)^2}{|D_{H_1}|}\right)\left(\frac{\delta - 3(\ell_H + N \cdot \ell_S + 1)^2/|D_{H_1}|}{\ell_H + \ell_S} - \frac{1}{|D_{H_2}|}\right) \approx \frac{\delta^2}{2(\ell_H + \ell_S)}, \tag{11}$$

where D_{H_1} and D_{H_2} define the ranges of random oracles H_1 and H_2, respectively.

Proof of Theorem 1. To prove the security of our IBMS scheme, we consider the following algorithm $\mathcal{B}$. On input $h_1, h_2, \cdots, h_{\ell_H+\ell_S} \in \{\mathbf{e}|\mathbf{e} \in \{-1, 0, 1\}^n, \|\mathbf{e}\|_1 \le \lambda\}$, algorithm $\mathcal{B}$ runs the forger $\mathcal{F}$ as follows.

(1) **Initialization**. $\mathcal{F}$ submits ID^* (the identity of the honest signer) to $\mathcal{B}$.
(2) **Setup**. Given the system parameter 1^n, $\mathcal{B}$ chooses a polynomial $h \in \mathcal{R}_q$ at random, sets $params = \langle h, H_0, H_1, H_2\rangle$ as the public parameters and returns *params* to $\mathcal{F}$, where H_0, H_1, and H_2 are hash functions controlled by $\mathcal{B}$.
(3) **Queries**. The algorithm $\mathcal{B}$ initializes counters $ctr1$, $ctr2$ to zero, and defines four initially empty lists $L_0[\cdot]$, $L_1[\cdot]$, $L_2[\cdot,\cdot]$ and $L_3[\cdot]$. List L_0, L_1 and L_2 simulate random oracles H_0, H_1 and H_2, while L_3 assigns a unique index $1 \le j \le \ell_H + N \cdot \ell_S$ to each ID_i either as a cosigner's identity in signing query, or as the first item in the argument of H_2 Query. Besides, $\mathcal{B}$ assigns index 0 to the honest signer ID^* by setting $L_3[ID^*] \leftarrow 0$, randomly chooses $s_1, s_2 \in \mathcal{D}_{\mathbb{Z}^n,s}$ satisfying $\|(s_1, s_2)\| \le s\sqrt{2n}$ and sets $S^* = (s_1, s_2)$. It then computes $P^* = s_1 + h \cdot s_2$ and sets $L_0[ID^*] \leftarrow (P^*, S^*)$. Finally, it responds to oracle queries as follows.
   (a) **$\mathbf{H_0}$-query**. When $\mathcal{F}$ queries the oracle H_0 with ID_i, $\mathcal{B}$ responds to $\mathcal{F}$ with P_i as follows.
      (i) If $L_0[ID_i]$ is defined, $\mathcal{B}$ parses $L_0[ID_i]$ as (P_i, S_i) and responds to $\mathcal{F}$ with P_i.
      (ii) Otherwise, $\mathcal{B}$ randomly chooses $s_{i,1}, s_{i,2} \in \mathcal{D}_{\mathbb{Z}^n,s}$ satisfying $\left\|(s_{i,1}, s_{i,2})\right\| \le s\sqrt{2n}$, computes the polynomial $P_i = s_{i,1} + h \cdot s_{i,2}$ and sets $L_0[ID_i] \leftarrow (P_i, S_i = (s_{i,1}, s_{i,2}))$. Finally, $\mathcal{B}$ responds to $\mathcal{F}$ with P_i.
   (b) **$\mathbf{H_1}$-query**. When $\mathcal{F}$ queries the oracle H_1 with r_i, $\mathcal{B}$ responds to $\mathcal{F}$ with d_i as follows.
      (i) If $L_1[r_i]$ is defined, $\mathcal{B}$ responds to $\mathcal{F}$ with $L_1[r_i]$.
      (ii) Otherwise, $\mathcal{B}$ randomly chooses $d_i \in \mathbb{Z}_q^n$ and sets $L_1[r_i] \leftarrow d_i$. Finally, $\mathcal{B}$ responds to $\mathcal{F}$ with d_i.
   (c) **$\mathbf{H_2}$-query**. When $\mathcal{F}$ queries the oracle H_2 with (ID_i, Q) where $Q = (r, \mathbf{ID}, m)$, $\mathcal{B}$ responds to $\mathcal{F}$ with c_i as follows.
      (i) If $L_3[ID_i]$ is undefined, $\mathcal{B}$ increases $ctr2$ and sets $L_3[ID_i] = ctr2$.
      (ii) Let $index = L_3[ID_i]$. If $L_2[index, Q]$ is defined, $\mathcal{B}$ responds to $\mathcal{F}$ with $L_2[index, Q]$.
      (iii) Otherwise, $\mathcal{B}$ randomly chooses c_j from the set $\{\mathbf{e}|\mathbf{e} \in \{-1, 0, 1\}^n, \|\mathbf{e}\|_1 \le \lambda\}$, sets $L_2[j, Q] \leftarrow c_j$ for $1 \le j \le \ell_H + N \cdot \ell_S$, and increases $ctr1$. Then $\mathcal{B}$ assigns $L_2[0, Q] = h_{ctr1}$ and responds to $\mathcal{F}$ with $L_2[index, Q]$.
   (d) **MExtract-query**. When $\mathcal{F}$ queries a signer's secret key with ID_i, $\mathcal{B}$ first checks whether the secret key S_i associated with ID_i has already presented in list L_0. If so, $\mathcal{B}$ responds the secret key S_i to $\mathcal{F}$. Otherwise, $\mathcal{B}$ issues the $H_0(ID_i)$ query to obtain the secret key S_i and responds S_i to $\mathcal{F}$.
   (e) **MSign-query**. When $\mathcal{F}$ queries a multi-signature with an identity set **ID** and a message m, $\mathcal{B}$ responds to $\mathcal{F}$ as follows.
      (i) $\mathcal{B}$ first checks whether $ID^* \in \mathbf{ID}$ or not. If not, it outputs $\perp$ and stops the protocol. It parses $\mathbf{ID} = \{ID_1 = ID^*, ID_2, \cdots, ID_N\}$, and checks whether $L_3[ID_k]$ has been defined for all $2 \le k \le N$. If not, it increases $ctr2$ and sets $L_3[ID_k] = ctr2$. Next, $\mathcal{B}$ increases counter $ctr1$ and sets $c_1 = h_{ctr1}$. Subsequently, $\mathcal{B}$ randomly chooses $z_{i,1}, z_{i,2} \in \mathcal{D}_{\mathbb{Z}^n,\sigma}$, computes $r_1 = \mathbf{A}z_1 - H_0(ID_1) \cdot c_1$, and queries $d_1 = H_1(r_1)$. If there exists more than one P_k in list L_0 satisfying $H_0(ID_k) = H_0(ID'_k) = P_k$ for $ID_k \ne ID'_k$, the event $\mathbf{bad_0}$ occurs and $\mathcal{B}$ stops the protocol by returning $\perp$. Finally, $\mathcal{B}$ broadcasts d_1 to other signers.
      (ii) After receiving d_k from the forger $\mathcal{F}$ (who plays the role of other signers) for $2 \le k \le N$, $\mathcal{B}$ checks whether $L_1[r_k] = d_k$ for each d_k. If any d_k can not be found in $L_1[r_k]$, $\mathcal{B}$ sets $alert \leftarrow true$ and returns r_1 to other signers.

On the other hand, if there exists more than one d_k in list L_1 satisfying $H_1(r_k) = H_1(r'_k) = d_k$ for $r_k \neq r'_k$, the event $\mathbf{bad_1}$ occurs and $\mathcal{B}$ aborts the protocol by returning $\perp$. Otherwise, $\mathcal{B}$ sets $r = \sum_{k=1}^{N} r_k \bmod q$ and checks whether $L_2[0, r, \mathbf{ID}, m]$ has already been defined. If so, the event $\mathbf{bad_2}$ occurs and $\mathcal{B}$ stops the protocol by returning $\perp$. If not, it sets $L_2[0, r, \mathbf{ID}, m] = c_1$, chooses random values from the set $\{\mathbf{e} | \mathbf{e} \in \{-1, 0, 1\}^n, \|\mathbf{e}\|_1 \leq \lambda\}$ and saves them into $L_2[j, r, \mathbf{ID}, m]$ for $1 \leq j \leq \ell_H + N \cdot \ell_S$. Subsequently, $\mathcal{B}$ broadcasts r_1 to other signers.

(iii) After receiving r_k from the forger $\mathcal{F}$ for $2 \leq k \leq N$, $\mathcal{B}$ checks whether $d_k = H_1(r_k)$ for each r_k. If there is any r_k that does not satisfy the equation, $\mathcal{B}$ aborts the protocol by returning $\perp$. If all r_k satisfy $H_1(r_k) = d_k$ for $2 \leq k \leq N$ and $alert = true$, the event $\mathbf{bad_3}$ occurs and $\mathcal{B}$ aborts the protocol by returning $\perp$. Otherwise, it broadcasts z_1 to other signers.

(iv) After receiving z_k from the forger $\mathcal{F}$ for $2 \leq k \leq N$, $\mathcal{B}$ computes $z = \sum_{k=1}^{N} z_k$ and responds to $\mathcal{F}$ with the multi-signature $\theta = (z, r)$.

(4) **Forgery**. With non-negligible probability, the forger $\mathcal{F}$ forges a valid multi-signature (z', r') with a message m' and the identity set $\mathbf{ID}'$. Note that $ID^* \in \mathbf{ID}'$, ID^* has not been issued during the **MExtract-query** phase and $(\mathbf{ID}', m')$ has not been issued during the **MSign-query** phase.

Assume that J is the index satisfying $1 \leq J \leq \ell_H + \ell_S$ and $L_2[0, r, \mathbf{ID}, m] = h_J$. We have the first tuple (z', r', h'_J) since $\mathcal{F}$ successfully forges a valid multi-signature, and we could obtain another tuple $(\hat{z}, \hat{r}, \widehat{h_J})$ by using the forking lemma [7]. According to the lemma, the environments of $\mathcal{F}$ provided by $\mathcal{B}$ are identical until the first call of $H_2(ID^*, r', \mathbf{ID}', m')$ or $H_2(ID^*, \hat{r}, \widehat{\mathbf{ID}}, \hat{m})$ leading to $L_2[0, r', \mathbf{ID}', m'] = h'_J$ and $L_2[0, \hat{r}, \widehat{\mathbf{ID}}, \hat{m}] = \widehat{h_J}$. Therefore, we have $m' = \hat{m}$, $\mathbf{ID}' = \widehat{\mathbf{ID}}$ and $r' = \hat{r}$. Furthermore, let I^* be the number of occurrences of ID^* in $\mathbf{ID}'$. Then all entries $L_3[ID'_i]$ and $c'_i = L_2[L_3[ID'_i], r', \mathbf{ID}', m']$ are initialized before the assignment $L_2[0, r', \mathbf{ID}', m'] = h'_J$. Consequently, we get the same hash values $c'_i = \widehat{c_i}$ in both executions for $ID'_i \neq ID^*$. On the other hand, we have $c'^* = h'_J$ and $\hat{c}^* = \widehat{h_J}$ for ID^* where $h'_J \neq \widehat{h_J}$. From $\mathbf{A}z' - \sum_{i=1}^{N} H_0(ID'_i)c'_i = \hat{r} = r' = \mathbf{A}\hat{z} - \sum_{i=1}^{N} H_0(ID'_i)\widehat{c_i}$ we could obtain

$$\mathbf{A}(z' - \hat{z} + |I^*| \cdot S^*(\hat{c}^* - c'^*)) = 0. \tag{12}$$

Owing to $\|z'\|, \|\hat{z}\| \leq 2N \cdot \sigma\sqrt{2n}$ and $\|S^*c'^*\|, \|S^*\hat{c}^*\| \leq \lambda \cdot s\sqrt{2n}$, we have $\|z' - \hat{z} + |I^*| \cdot S^*(\hat{c}^* - c'^*)\| \leq 4N \cdot \sigma\sqrt{2n} + 2|I^*| \cdot \lambda s\sqrt{2n}$. Using $\mathbf{A} = (\mathbf{1}, h)$ and $S^* = (s_1, s_2)$ we could obtain the following equality

$$(z'^{(1)} - \hat{z}^{(1)} + |I^*| \cdot s_1(\hat{c}^* - c'^*)) + h \cdot (z'^{(2)} - \hat{z}^{(2)} + |I^*| \cdot s_2(\hat{c}^* - c'^*)) = 0. \tag{13}$$

According to Eq. (13) and [19,33], we could obtain two polynomials $\mathbf{u}_1$ and $\mathbf{u}_2$ with small coefficients satisfying $\mathbf{u}_1 + h \cdot \mathbf{u}_2 = 0$. In other words, we solve Ring-SIS$_{2,q,\beta}$ for $\beta \leq 4N \cdot \sigma\sqrt{2n} + 2|I^*| \cdot \lambda s\sqrt{2n}$, which is presumed to be hard. Note that the number of signers in the signing process $N \leq N_{max}$ should not be too large.

Suppose that the forger $\mathcal{F}$ can successfully forge a valid multi-signature with probability δ. We can bound the winning probability acc of $\mathcal{B}$ as follows.

$$\begin{aligned} acc &\geq \delta - \sum_{i=0}^{3} \Pr[\mathbf{bad}_i] \\ &\geq \delta - \frac{(\ell_H + \ell_S + 1)^2}{|D_{H_0}|} - \frac{(\ell_H + N \cdot \ell_S + 1)^2}{|D_{H_1}|} - \sum_{i=1}^{\ell_S} \left(\frac{(\ell_H + N \cdot \ell_S)}{|D_{H_1}|} + \frac{(\ell_H + \ell_S)}{|D_{H_1}|} + \frac{N}{|D_{H_1}|} \right) \\ &\geq \delta - \frac{(\ell_H + \ell_S + 1)^2}{|D_{H_0}|} - \frac{(\ell_H + N \cdot \ell_S + 1)^2}{|D_{H_1}|} - \frac{\ell_S(\ell_H + N \cdot \ell_S)}{|D_{H_1}|} - \frac{\ell_S(\ell_H + \ell_S + N)}{|D_{H_1}|} \\ &\geq \delta - \frac{3(\ell_H + N \cdot \ell_S + 1)^2}{|D_{H_1}|} \end{aligned} \tag{14}$$

After using the forking lemma [7], the probability frk is

$$\begin{aligned} frk &\geq acc \cdot \left(\frac{acc}{\ell_H + \ell_S} - \frac{1}{|D_{H_2}|} \right) \\ &\geq \left(\delta - \frac{3(\ell_H + N \cdot \ell_S + 1)^2}{|D_{H_1}|} \right) \cdot \left(\frac{\delta - 3(\ell_H + N \cdot \ell_S + 1)^2/|D_{H_1}|}{\ell_H + \ell_S} - \frac{1}{|D_{H_2}|} \right). \end{aligned} \tag{15}$$

Finally, the algorithm $\mathcal{B}$ can solve Ring-SIS$_{2,q,\beta}$ with probability sol at least

$$\left(\frac{1}{2} - 2^{-100} \right) \left(\delta - \frac{3(\ell_H + N \cdot \ell_S + 1)^2}{|D_{H_1}|} \right) \left(\frac{\delta - 3(\ell_H + N \cdot \ell_S + 1)^2/|D_{H_1}|}{\ell_H + \ell_S} - \frac{1}{|D_{H_2}|} \right) \approx \frac{\delta^2}{2(\ell_H + \ell_S)}. \tag{16}$$

If δ is non-negligible, so is sol. Therefore, our scheme is existentially unforgeable under selective identity and adaptive chosen message attacks (*EUF-SID-CMA* secure). □

Table 1
Comparison among RIBS scheme, BS scheme, MSig scheme, MS scheme, PLMS scheme and our IBMS scheme.

Schemes	Secret key size	Multi-signature size	Computational cost		ID-based	Security
			Sign	Verify		
RIBS [23]	$\leq 4n\log(s\sqrt{n})$	$2Nn\log 12\sigma + N\cdot\lambda(\log n+1)$	$3T_{pm}$	$2T_{pm}$	Yes	EUF-ID-CMA
BS [5]	$2n\log(2d+1)$	$2n\log(2N(k-d\cdot 32)+1)+n\log q$	$2T_{pm}$	$(N+1)T_{pm}$	No	SUF-CMA
MSig [12]	$2n\log(2d+1)$	$2n\log(2N(n-1)\sqrt{n}d\log^2 n+1)+n\log q$	$2T_{pm}$	$(N+1)T_{pm}$	No	EUF-CMA
MS [26]	$2n\log(2d+1)$	$2n\log(2N(k-d\cdot 32)+1)+2n\log q$	$(N+2)T_{pm}$	$2T_{pm}$	No	EUF-CMA
PLMS [35]	$2n\log(2d+1)$	$2n\log(2N(k-d\cdot 32)+1)+160N$	$2T_{pm}$	$(N+1)T_{pm}$	No	EUF-CMA
Our IBMS	$\leq 2n\log(s\sqrt{n})$	$2n\log(N\cdot 12\sigma)+n\log q$	$2T_{pm}$	$(N+1)T_{pm}$	Yes	EUF-SID-CMA

Legends: n: the system parameter; k, d, λ: positive integers; N: the number of cosigners; $s = \left\|\widetilde{\mathbf{B}_{f,g}}\right\| \omega(\sqrt{\log 2n})$; $\sigma = 12\lambda s\sqrt{n}$; T_{pm}: the cost of executing a polynomial multiplication operation.

Table 2
Comparison between PLMS scheme [35] and our IBMS scheme.

Schemes	Sizes (KB)			Root Hermite factor γ	Security parameter	Parameter set
	$\|pk\|$	$\|sk\|$	$\|\theta\|$			
PLMS [35]	1.69	0.20	2.51	1.007	80	SetI
IBMS	$\|ID\|$	0.20	3.60			
PLMS [35]	1.81	0.20	2.74	1.007	80	SetII
IBMS	$\|ID\|$	0.20	3.85			
PLMS [35]	3.49	0.40	5.18	1.004	192	SetIII
IBMS	$\|ID\|$	0.40	7.82			
PLMS [35]	3.63	0.40	5.53	1.004	192	SetIV
IBMS	$\|ID\|$	0.40	8.20			

Legends: $|ID|$: the size of an identity of a signer; $|\theta|$: the size of multi-signature.

6. Comparison

As shown in Table 1, we make the performance analysis by presenting the comparison among RIBS scheme [23], BS scheme [5], MSig scheme [12], MS scheme [26], PLMS scheme [35] and our IBMS scheme, in terms of the sizes of signing key and multi-signature, computational costs of signing algorithm and verification algorithm, identity-based functionality and security property.

Compared with the RIBS scheme [23] which is an identity-based signature (not multi-signature) scheme over lattice, our IBMS scheme supports the multi-signature functionality. Signature size in our scheme is much smaller than a bundled signature consisting of N single signatures (of RIBS scheme), as we perform an accumulation operation on N signatures instead of simply concatenating them. Compared with the existing multi-signature schemes [5,12,26,35] which work in the PKI setting, our scheme implements identity-based functionality. Namely, our scheme enables each signer to select its system identity information (such as email address, physical IP address and etc.) as the public key, which effectively alleviates the certificate management problem in the PKI setting. Furthermore, our scheme takes the advantage of discrete Gaussian distribution instead of uniform distribution to generate secret signing keys and multi-signatures. Using the discrete Gaussian distribution, we could take the Gaussian sampling algorithm to generate users' signing keys corresponding to their identities without leaking any information about the trapdoor. Unlike [5,12,26,35], our scheme is a pure identity-based cryptosystem and does not require all the system users to share a trusted common string, which further simplifies the deployment of our scheme in practice. On the other hand, the PLMS scheme [35] is the most efficient among the four above-mentioned PKI-based multi-signature schemes [5,12,26,35]. Table 2 shows the concrete parameter comparison between PLMS scheme [35] and our IBMS scheme. To be more intuitive, we make the multi-signature size comparison between two schemes in Fig. 3.

According to [11], the root Hermite factor γ is important to analyze the security of signature schemes. For $\gamma \approx 1.007$, solving the lattice problem is at least 2^{80}-hard. The problem seems to be much more difficult and is at least 2^{192}-hard approximately when $\gamma \approx 1.004$. As shown in Table 2 and Fig. 3, although it has slightly larger multi-signature size, our IBMS scheme implements identity-based functionality in multi-signature, enabling each signer to select its system identity information (such as email address, physical IP address and etc.) as the public key and alleviating the certificate management problem in the PKI setting effectively. Meanwhile, our scheme has similar security level and computational cost as the schemes in comparison.

In Table 3 and Table 4, we propose four sets of parameters for PLMS scheme [35] and our IBMS scheme for the comparison, respectively.

Concretely, we compute the size of public key, private key, and multi-signature in the PLMS scheme [35] as follows. We take the parameter setI from Table 3 as an example. The public key size is $n\lceil\log q\rceil = 13824$ bits (≈ 1.69 KB) since the

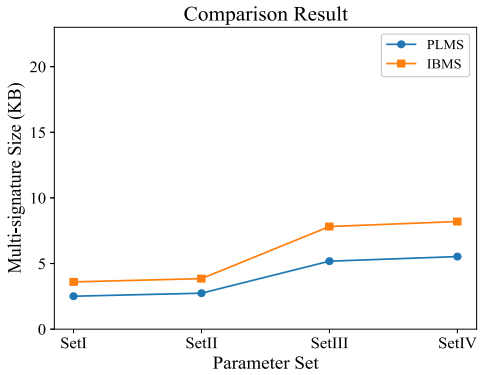

Fig. 3. Multi-signature size comparison.

Table 3
Four parameter sets of PLMS scheme [35].

Parameters	SetI	SetII	SetIII	SetIV
Lattice dimension n	512	512	1024	1024
Modulus q	134136833	536856577	258001921	536856577
Group signers N	5	10	5	10
Range k	65540	65540	131040	131040
Range d	1	1	1	1
Private key bit size	1623	1623	3246	3246
Public key bit size	13824	14848	28613	29696
Multi-signature bit size $\|\theta\|$	20585	22410	42418	45266
Root Hermite factor γ	1.007	1.007	1.004	1.004

Table 4
Four parameter sets of our IBMS scheme.

Parameters	SetI	SetII	SetIII	SetIV
Lattice dimension n	512	512	1024	1024
Modulus q	2^{22}	2^{24}	2^{24}	2^{25}
Group signers N	5	10	5	10
Positive number λ	14	14	28	28
Range d	1	1	1	1
Private key bit size	1623	1623	3246	3246
Multi-signature bit size $\|\theta\|$	29491	31539	64101	67173
Root Hermite factor γ	1.007	1.007	1.004	1.004

public key $\mathbf{T}_i$ is in $\mathcal{R}^q$. Consisting of two polynomials $s_i^{(1)}, s_i^{(2)}$ from $\mathcal{R}_1^q$, the size of the private key $\mathbf{sk}_i$ is $2n\lceil\log 3\rceil = 1623$ bits (≈ 0.20 KB). The multi-signature consists of polynomials $\mathbf{z}$ and $\{\mathbf{c}_1, \cdots, \mathbf{c}_N\}$. Notice that $\mathbf{z}$ is in $\mathcal{R}^q_{N\cdot(k-32)} \times \mathcal{R}^q_{N\cdot(k-32)}$, so its size is $2n\lceil\log(2N\cdot(k-32)+1)\rceil = 19785$ bits. In addition, the polynomials $\{\mathbf{c}_1, \cdots, \mathbf{c}_N\}$ have size of 800 bits as each $\mathbf{c}_i$ is 160 bits in the PLMS scheme. Therefore, we can obtain that the multi-signature size of [35] is 20585 bits ($\approx$ 2.51 KB). Finally, we can compute the concrete root Hermite factor $\gamma = (\beta/\sqrt{q})^{\frac{1}{2n}} \approx 1.007$.

Next, we compute the size of private key and multi-signature in our IBMS scheme as follows while the public key (denoted by $|ID|$) is selected from a user's identity. We take the parameter setI from Table 4 as an example. The private key S_i consists of two polynomials $s_{i,1}, s_{i,2}$ sampled from the set $\{-1, 0, 1\}$, so the size of private key is $2n\lceil\log 3\rceil = 1623$ bits (≈ 0.20 KB). The multi-signature consists of polynomials z and r, where the size of z is $2n\lceil\log(N\cdot 12\sigma)\rceil = 18227$ bits. In addition, the polynomial r in $\mathcal{R}_q$ has a size of $n\lceil\log q\rceil = 11264$ bits. Therefore, we can obtain the multi-signature size of our scheme is 29491 bits ($\approx$ 3.60 KB). Finally, we can compute the concrete root Hermite factor $\gamma = (\beta/\sqrt{q})^{\frac{1}{2n}} \approx 1.007$.

7. Conclusion

In this paper we proposed an identity-based multi-signature (IBMS) scheme over NTRU lattices which is provably secure in the random oracle model based on Ring-SIS assumption. To the best of our knowledge, it is the first IBMS scheme over NTRU lattices in the literature. Compared with the RIBS scheme [23] which is an identity-based signature (not multi-signature) scheme over lattice, our IBMS scheme supports the multi-signature functionality. Signature size in our scheme is much smaller compared to a bundled signature consisting of N single signatures (of RIBS), as we perform an accumulation operation on N signatures instead of simply concatenating them. Compared with the existing PKI-based multi-signature

schemes [5,12,26,35], our scheme implements identity-based functionality, allowing each signer to use its identity information as the public key, which alleviates the certificate management problem in the PKI setting effectively. It also takes the advantage of discrete Gaussian distribution instead of uniform distribution to generate secret signing keys and multi-signatures to refrain from leaking any information about the trapdoor. Besides, our scheme does not require all the system users to share a trusted common string, which further simplifies the deployment of our scheme in practice. As a future work, we will consider constructing an IBMS scheme with shorter multi-signature size to efficiently reduce the consumption of storage space and transmission bandwidth.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] M. Ajtai, Generating hard instances of lattice problems (extended abstract), in: Proceedings of the Twenty-Eighth Annual ACM Symposium on Theory of Computing - STOC'96, ACM Press, 1996, pp. 99–108.
[2] A.R. Babu, N.B. Gayathri, P.V. Reddy, Efficient ID-based key-insulated multi signature scheme without pairings, in: 2019 Innovations in Power and Advanced Computing Technologies (i-PACT), IEEE, mar 2019.
[3] A. Bagherzandi, J.-H. Cheon, S. Jarecki, Multisignatures secure under the discrete logarithm assumption and a generalized forking lemma, in: Proceedings of the 15th ACM Conference on Computer and Communications Security - CCS'08, ACM Press, 2008, pp. 449–458.
[4] A. Bagherzandi, S. Jarecki, Identity-based aggregate and multi-signature schemes based on RSA, in: Public Key Cryptography – PKC 2010, Springer, Berlin, Heidelberg, 2010, pp. 480–498.
[5] R.E. Bansarkhani, J. Sturm, An efficient lattice-based multisignature scheme with applications to bitcoins, in: Cryptology and Network Security, Springer International Publishing, 2016, pp. 140–155.
[6] M. Bellare, G. Neven, Identity-based multi-signatures from RSA, in: Lecture Notes in Computer Science, Springer, Berlin, Heidelberg, 2006, pp. 145–162.
[7] M. Bellare, G. Neven, Multi-signatures in the plain public-key model and a general forking lemma, in: Proceedings of the 13th ACM Conference on Computer and Communications Security - CCS'06, ACM Press, 2006, pp. 390–399.
[8] D. Boneh, M. Drijvers, G. Neven, Compact multi-signatures for smaller blockchains, in: Lecture Notes in Computer Science, Springer International Publishing, 2018, pp. 435–464.
[9] R. Choi, K. Kim, Lattice-based multi-signature with linear homomorphism, in: 2016 Symposium on Cryptography and Information Security, The Institute of Electronics, Information and Communication Engineers, Tokyo, Japan, 2016.
[10] W. Diffie, M. Hellman, New directions in cryptography, IEEE Trans. Inf. Theory 22 (6) (nov 1976) 644–654.
[11] L. Ducas, V. Lyubashevsky, T. Prest, Efficient identity-based encryption over NTRU lattices, in: Lecture Notes in Computer Science, Springer, Berlin, Heidelberg, 2014, pp. 22–41.
[12] M. Fukumitsu, S. Hasegawa, A tightly-secure lattice-based multisignature, in: Proceedings of the 6th on ASIA Public-Key Cryptography Workshop - APKC'19, ACM Press, 2019, pp. 3–11.
[13] C. Gentry, C. Peikert, V. Vaikuntanathan, Trapdoors for hard lattices and new cryptographic constructions, in: Proceedings of the Fortieth Annual ACM Symposium on Theory of Computing - STOC 08, ACM Press, 2008, pp. 197–206.
[14] S. Goldwasser, S. Micali, R.L. Rivest, A digital signature scheme secure against adaptive chosen-message attacks, SIAM J. Comput. 17 (2) (apr 1988) 281–308.
[15] L.K. Grover, Quantum mechanics helps in searching for a needle in a haystack, Phys. Rev. Lett. 79 (2) (jul 1997) 325–328.
[16] H. Guo, W. Li, E. Meamari, C.-C. Shen, M. Nejad, Attribute-based multi-signature and encryption for EHR management: a blockchain-based solution, in: 2020 IEEE International Conference on Blockchain and Cryptocurrency (ICBC), IEEE, may 2020.
[17] J. Guo, W. Wu, Adaptive influence maximization: if influential node unwilling to be the seed, ACM Trans. Knowl. Discov. Data 15 (5) (jun 2021) 1–23.
[18] J. Guo, W. Wu, Continuous profit maximization: a study of unconstrained dr-submodular maximization, IEEE Trans. Comput. Soc. Syst. 8 (3) (jun 2021) 768–779.
[19] T. Güneysu, V. Lyubashevsky, T. Pöppelmann, Practical lattice-based cryptography: a signature scheme for embedded systems, in: Cryptographic Hardware and Embedded Systems – CHES 2012, Springer, Berlin, Heidelberg, 2012, pp. 530–547.
[20] L. Harn, C.-H. Lin, Contract signature in e-commerce, Comput. Electr. Eng. 37 (2) (mar 2011) 169–173.
[21] J. Hoffstein, N. Howgrave-Graham, J. Pipher, J.H. Silverman, W. Whyte, NTRUSign: digital signatures using the NTRU lattice, in: Topics in Cryptology — CT-RSA 2003, Springer, Berlin, Heidelberg, 2003, pp. 122–140.
[22] J. Hoffstein, J. Pipher, J.H. Silverman, NTRU: a ring-based public key cryptosystem, in: Lecture Notes in Computer Science, Springer, Berlin, Heidelberg, 1998, pp. 267–288.
[23] Y.-H. Hung, Y.-M. Tseng, S.-S. Huang, Revocable ID-based signature with short size over lattices, Secur. Commun. Netw. 2017 (2017) 1–9.
[24] S.H. Islam, A.K. Das, M.K. Khan, Design of a provably secure identity-based digital multi-signature scheme using biometrics and fuzzy extractor, Secur. Commun. Netw. 9 (16) (jul 2016) 3229–3238.
[25] K. Itakura, K. Nakamura, A public-key cryptosystem suitable for digital multisignatures, NEC Res. Development 71 (1983) 1–8.
[26] M. Kansal, A.K. Singh, R. Dutta, Efficient multi-signature scheme using lattice, Comput. J. (may 2021).
[27] F.Y. Kong, L.H. Diao, J. Yu, Y.L. Jiang, D.S. Zhou, Lattice-based multi-signature schemes, Appl. Mech. Mater. 411–414 (sep 2013) 3–6.
[28] Q. Lei, Sequential-multi digital signature application in electronic contract signing, Comput. Modern. (2) (2016) 72–74, 81.
[29] C.-m. Li, L.-j. Chen, R. Shi, Design and implement of EMR XML signature schema, Chongqing Med. 38 (21) (2009) 2651–2652.
[30] C.-M. Li, T. Hwang, N.-Y. Lee, Threshold-multisignature schemes where suspected forgery implies traceability of adversarial shareholders, in: Advances in Cryptology — EUROCRYPT'94, Springer, Berlin, Heidelberg, 1995, pp. 194–204.
[31] X. Lu, W. Yin, Q. Wen, Z. Jin, W. Li, A lattice-based unordered aggregate signature scheme based on the intersection method, IEEE Access 6 (2018) 33986–33994.
[32] A. Lysyanskaya, Unique signatures and verifiable random functions from the DH-DDH separation, in: Advances in Cryptology — CRYPTO 2002, Springer, Berlin, Heidelberg, 2002, pp. 597–612.
[33] V. Lyubashevsky, Lattice signatures without trapdoors, in: Advances in Cryptology – EUROCRYPT 2012, Springer, Berlin, Heidelberg, 2012, pp. 738–755.
[34] V. Lyubashevsky, C. Peikert, O. Regev, On ideal lattices and learning with errors over rings, in: Advances in Cryptology – EUROCRYPT 2010, Springer, Berlin, Heidelberg, 2010, pp. 1–23.

[35] C. Ma, M. Jiang, Practical lattice-based multisignature schemes for blockchains, IEEE Access 7 (2019) 179765–179778.
[36] C. Ma, J. Weng, Y. Li, R. Deng, Efficient discrete logarithm based multi-signature scheme in the plain public key model, Des. Codes Cryptogr. 54 (2) (jul 2010) 121–133.
[37] G. Maxwell, A. Poelstra, Y. Seurin, P. Wuille, Simple Schnorr multi-signatures with applications to bitcoin, Des. Codes Cryptogr. 87 (9) (feb 2019) 2139–2164.
[38] S. Micali, K. Ohta, L. Reyzin, Accountable-subgroup multisignatures, in: Proceedings of the 8th ACM Conference on Computer and Communications Security - CCS'01, ACM Press, 2001, pp. 245–254.
[39] D. Micciancio, S.P. Vadhan, Statistical zero-knowledge proofs with efficient provers: lattice problems and more, in: Advances in Cryptology - CRYPTO 2003, Springer, Berlin, Heidelberg, 2003, pp. 282–298.
[40] K. Ohta, T. Okamoto, Multi-signature schemes secure against active insider attacks, IEICE Trans. Fundam. Electron. Commun. Comput. Sci. 82 (1999) 21–31.
[41] C. Peng, X. Du, New lattice-based digital multi-signature scheme, in: Communications in Computer and Information Science, Springer, Singapore, 2020, pp. 129–137.
[42] E.D. Raj, G. Manogaran, G. Srivastava, Y. Wu, Information granulation-based community detection for social networks, IEEE Trans. Comput. Soc. Syst. 8 (1) (feb 2021) 122–133.
[43] A. Shamir, Identity-based cryptosystems and signature schemes, in: Advances in Cryptology, Springer, Berlin, Heidelberg, 1984, pp. 47–53.
[44] P. Shor, Algorithms for quantum computation: discrete logarithms and factoring, in: Proceedings 35th Annual Symposium on Foundations of Computer Science, IEEE Comput. Soc. Press, 1994, pp. 124–134.
[45] D. Stehlé, R. Steinfeld, Making NTRU as secure as worst-case problems over ideal lattices, in: Advances in Cryptology – EUROCRYPT 2011, Springer, Berlin, Heidelberg, 2011, pp. 27–47.
[46] E. Syta, I. Tamas, D. Visher, D.I. Wolinsky, P. Jovanovic, L. Gasser, N. Gailly, I. Khoffi, B. Ford, Keeping authorities "honest or bust" with decentralized witness cosigning, in: 2016 IEEE Symposium on Security and Privacy (SP), IEEE, may 2016, pp. 526–545.
[47] C. Thirumalai, S. Mohan, G. Srivastava, An efficient public key secure scheme for cloud and IoT security, Comput. Commun. 150 (jan 2020) 634–643.
[48] M. Tian, L. Huang, Identity-based signatures from lattices: simpler, faster, shorter, Fundam. Inform. 145 (2016) 171–187.
[49] L. Wei, L. Zhang, D. Huang, K. Zhang, L. Dai, G. Wu, PSDAAP: provably secure data authenticated aggregation protocols using identity-based multi-signature in marine WSNs, Sensors 17 (9) (sep 2017) 2117.
[50] C. Xie, J. Weng, J. Weng, L. Hou, Scalable revocable identity-based signature over lattices in the standard model, Inf. Sci. 518 (may 2020) 29–38.